Resistive switching and memory computing

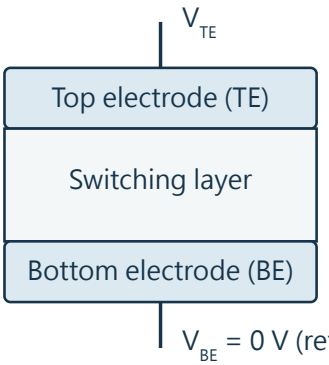
A resistive memory cell typically has a top electrode, switching layer and bottom electrode. A programming pulse changes its stored state; a specified read bias probes its current.

LRS and HRS mean low and high resistance. Assigning either state to a binary value is a system convention, not a material law. We connect measurements to physical models, arrays and reliability without treating one observation as proof of a unique mechanism.

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Cell structure and voltage reference

Two-terminal cell: define voltage first



Define $V = V_{TE} - V_{BE}$

Operations and readout

SET: higher conductance

RESET: lower conductance

Read: state voltage and duration

$G_{read} = I_{read} / V_{read}$

Polarity and current limit are device-specific; no particular mechanism is assumed.

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Read conductance has measurement conditions

At a specified voltage and time, define $G_{read} = I_{read}/V_{read}$. This ratio is an apparent conductance at that operating point.

Nonlinear response, polarity asymmetry or drift can make it different at another bias or time. A read can also disturb the stored state.

A voltage sweep traces a response under a specified sweep rate and current limit. Pulse experiments specify amplitude, duration and sequence and can probe switching and read disturbance.

Report temperature, voltage polarity, compliance, pulse timing and read protocol. A single curve does not establish device-to-device or cycle-to-cycle distributions.

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Forming, SET, RESET and read

- Forming is an initial conditioning step required by some devices, not all.
- SET generally moves to a more conductive state; RESET moves to a less conductive state.
- Read probes the state. Current compliance constrains programming current but does not guarantee freedom from damage.

For bipolar switching, SET and RESET generally use opposite bias polarities. Unipolar switching can use the same polarity with different pulse or current-limit conditions.

Define voltage explicitly, for example $V = V_{top} - V_{bottom}$, before naming positive or negative. A polarity observed in one stack is not universal.

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Trace a pulse protocol

This synthetic example assumes a particular bipolar stack and $V = V_{top} - V_{bottom}$; the numbers are illustrative, not specifications. Record the read immediately after each pulse and again later to detect drift.

Step	Applied pulse	Current limit	Read at 0.1 V	Inference
Initial read	+0.1 V, 10 µs	read circuit	1 µA	$G_{read} = 10$ µS
SET	+1.0 V, 1 µs	100 µA	20 µA	$G_{read} = 200$ µS
RESET	−1.0 V, 1 µs	not specified	1 µA	$G_{read} = 10$ µS

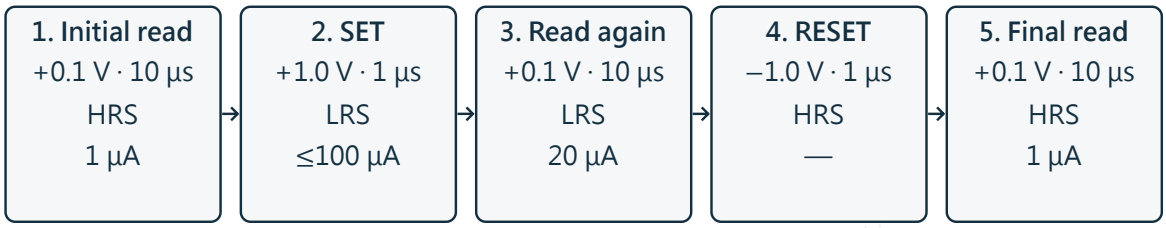
Compliance constrains programming current; a low read pulse is not automatically nondestructive. Verify repeated reads and record drift instead of inferring it from one measurement.

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Pulse operations and state sequence

Pulse and read sequence: illustrative example

Example only; V = Vtop − Vbottom.



Read currents shown; SET limited; RESET current unspecified.
Not to scale; reads assumed nondisturbing.

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ECM: a possible redox pathway

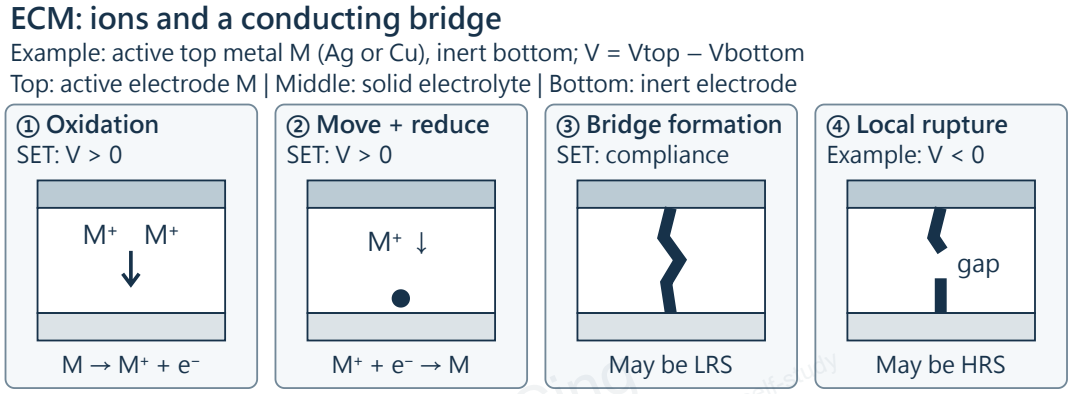
In electrochemical metallization (ECM), an active electrode can oxidize; metal cations migrate through the switching layer and reduction can form a conductive bridge. Subsequent operation can narrow or rupture it.

For a hypothetical Ag / oxide / inert-electrode stack with Ag positive during SET, oxidation $\mathrm{Ag} \rightarrow \mathrm{Ag}^{+} + e^{-}$ supplies cations while electrons enter the external circuit. Reduction $\mathrm{Ag}^{+} + e^{-} \rightarrow \mathrm{Ag}$ can deposit metal where electrons are available.

Nucleation, growth direction and RESET depend on electron supply, transport and reaction kinetics. This polarity and schematic are examples, not universal rules.

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ECM stages for one electrode configuration



Schematic, not to scale; growth direction depends on materials and reaction kinetics.
RESET polarity, rupture location and pulse depend on device; specify read conditions.

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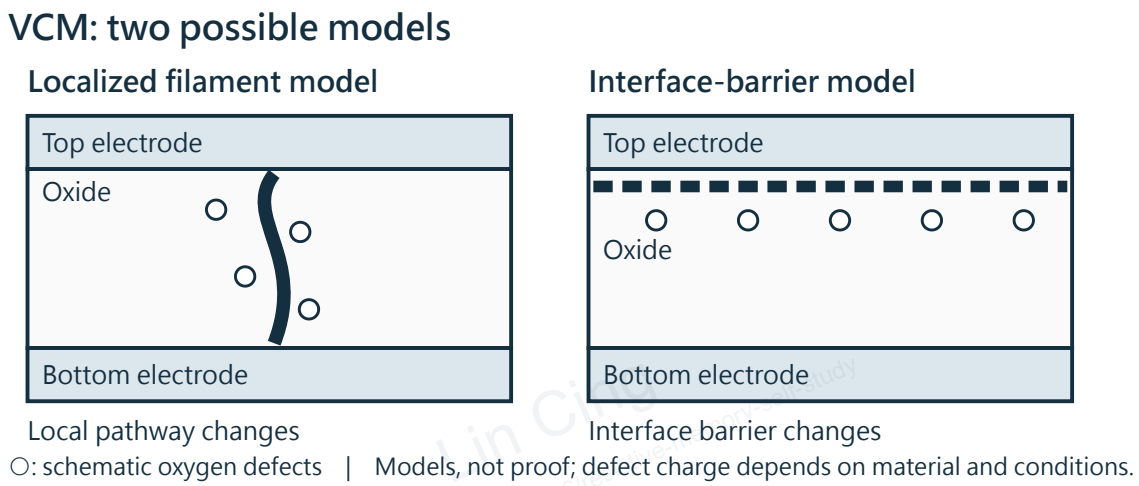
VCM and competing explanations

Valence-change mechanisms (VCM) involve oxygen-ion or oxygen-vacancy transport and local redox changes. Redistribution changes local stoichiometry and electronic transport. One model emphasizes a filament; another emphasizes an interface barrier. Each requires specific electrode, field and redox assumptions. Thermochemical contributions and unipolar behavior are not universally equivalent.

A hysteretic I–V curve is not proof of a unique mechanism. Local chemical maps, conductive microscopy, in-situ imaging and time-resolved measurements can test models, subject to resolution and possible measurement perturbations; correlation alone does not establish causality.

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Two candidate VCM models



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Model a state without assuming a mechanism

One framework is $i(t) = G(x(t), v(t), t)v(t)$ and $\dot{x} = f(x, v, t)$. The chosen state x may be multidimensional, and G need not be independent of voltage or time.

An ideal memristor is not synonymous with every two-terminal device displaying hysteresis. A model must state its domain and be tested against measurements.

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An ideal crossbar computes column currents

Let row i be driven at V_i and the conductance at row i, column j be G_{ij}. Assume linear read response, negligible wire resistance, and columns held at virtual ground.

Each branch carries $I_{ij} = G_{ij}V_i$. Kirchhoff's current law then gives $I_j = \sum_i G_{ij}V_i$. These assumptions are part of the result.

For $G = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}\ \mu\mathrm{S}$ and $V = (0.2, 0.1)\ \mathrm{V}$, the two column currents are $I_1 = 2(0.2) + 1(0.1) = 0.5\,\mu\mathrm{A}$ and $I_2 = 1(0.2) + 3(0.1) = 0.5\,\mu\mathrm{A}$. Here $\mu\mathrm{S} \cdot \mathrm{V} = \mu\mathrm{A}$; the numbers illustrate the model.

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Derive the array sum at the cell terminals

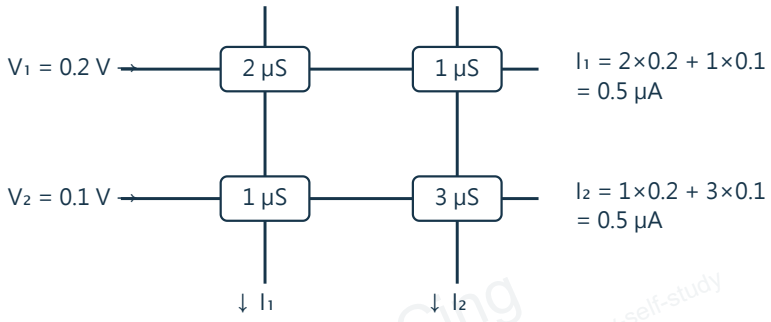
Let the local row and column potentials at cell (i, j) be v_{ij} and u_{ij}. For a linear read conductance, the branch current into the column is $I_{ij} = G_{ij}(v_{ij} - u_{ij})$. Kirchhoff's current law sums these branch currents at the column sensing node. Only when row wires are ideal and the column is held at virtual ground do $v_{ij} = V_i$ and $u_{ij} = 0$ yield $I_j = \sum_i G_{ij} V_i$.

With the earlier 2×2 example, suppose both local row-2 potentials are 0.08 V, row 1 remains 0.2 V, and column potentials remain zero. Then $I_1 = 2(0.2) + 1(0.08) = 0.48$ µA and $I_2 = 1(0.2) + 3(0.08) = 0.44$ µA. This assumes the local voltages are known; solving a real resistive network determines them rather than prescribing them.

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Ideal crossbar: two input rows

2 × 2 crossbar: sum branch currents by column



Assumptions: linear read, no wire resistance, virtual-ground columns.
$I_j = \Sigma_i G_{ij} V_i$; units: µS × V = µA.

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When the array differs from the model

Wire resistance changes local voltages; finite sensing impedance changes column potential. Leakage, unintended paths, nonlinear response, conductance drift and programming error can change the measured sum.

A negative weight needs an explicit representation, such as a difference of conductances, and corresponding sensing circuitry. A single passive conductance is not a negative weight.

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Reliability is a measurement, not a slogan

Retention describes distinguishability after a stated time and temperature, with a defined read and failure criterion. Endurance is a cycle count under stated waveforms and failure criterion.

Separate cycle-to-cycle from device-to-device variation. Read margin depends on the sensing circuit and distributions, not merely two nominal resistance values.

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Compare reliability only under matching protocols

Consider two hypothetical measurements, each on 100 cells. Test A reports 95 cells retaining distinguishable read currents after 1 hour at 85 °C; test B reports 98 cells after 1 hour at 25 °C. Both must specify read voltage and time, current thresholds, prior cycling and whether failed cells were excluded. Even with those records, different temperatures prevent treating the observed percentages as a controlled comparison of two materials.

For endurance, record the SET/RESET amplitudes, pulse widths, verify procedure and failure criterion. A short high-temperature retention measurement alone cannot establish 10-year retention at room temperature; an extrapolation requires a supported model, uncertainty and appropriate validation.

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Applications and device requirements

Four uses: operations and device demands

Storage	Write state → read and decide Focus: read margin, retention, write endurance
Stateful logic	Read state → update state Focus: operations may consume writes
Offline inference	Program weights → repeated reads Focus: read accuracy, drift and converters
Online training	Read and compute → update weights Focus: update error, asymmetry and endurance

Conceptual flows; operations and peripheral costs depend on architecture.

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Storage and selectors: the array sets the constraints

For digital storage, a read circuit must distinguish the programmed states at specified voltages and times. A large single-cell resistance ratio does not by itself guarantee a usable array read margin: wire resistance, sensing noise and currents through unselected cells also matter.

An access device can suppress unselected-path currents using gate control, as in 1T1R, or using a nonlinear two-terminal selector. Its leakage, drive current, switching speed and integration cost must be assessed together with the memory element; a selector is not automatically an ideal switch.

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Stateful logic can require repeated writes

In stateful logic, a programmed device state participates in a logic operation and the result is stored by a subsequent state change. The precise voltage sequence and number of writes depend on the circuit and its logic convention. A reconfigurable circuit instead may write a configuration once and use it for many operations; do not assume each operation rewrites its configuration.

Suppose the **same cell** receives exactly k writes per operation, operations occur at N per second, and this rate is constant for T seconds. The cell accumulates NkT writes. For $k = 2$, $N = 10^3\,\mathrm{s}^{-1}$ and $T = 60\,\mathrm{s}$, this is 1.2×10^5 writes. If writes are spread across cells, sum each cell's actual writes instead. A write is not necessarily a complete SET/RESET cycle; endurance depends on the specified pulses and failure criterion.

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Inference and training stress different properties

An analog crossbar can use programmed conductances to perform an approximate weighted sum. Offline inference programs weights before use: read noise, conductance drift, line resistance and ADC/DAC precision affect the result. Weight signs need an explicit representation, such as a differential pair.

Online training also updates conductances during learning. Update nonlinearity, asymmetry, quantization, stochasticity and write endurance can affect the training trajectory. Impact depends on the model, workload, mapping and compensation; neither inference nor training is automatically tolerant of every device imperfection.

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Temporal randomness differs from a device fingerprint

A stochastic switching device may produce different outcomes across repeated identical pulses. A random-number generator uses temporal variation; bias, serial correlation, readout and any post-processing must be characterized. Repeated switching also consumes endurance.

A physical unclonable function (PUF) instead aims to extract a reproducible device-specific response from spatial differences among devices. Its usefulness depends on stability under environmental changes, uniqueness across devices and measurement noise. Temporal randomness and spatial variation are different properties, even if a single device technology exhibits both.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Arrays, selectors and three-dimensional integration

A single RRAM cell must become part of an addressable network. This chapter follows the current through that network, then examines access devices, sensing and the physical layout.

All numerical examples are deliberately simplified circuit models, not measured device specifications.

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A crosspoint is a cell, not an isolation switch

At the intersection of row R_i and column C_j, a memory element connects the two lines. Applying a voltage to a selected row and sensing a column also establishes voltages across other intersections.

Those other cells can carry currents into the sense circuit, or form a path through floating lines. Which currents appear depends on the line-bias and sensing circuit; a cell's HRS/LRS ratio alone does not determine array performance.

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Fixed $V/2$ bias: define every line

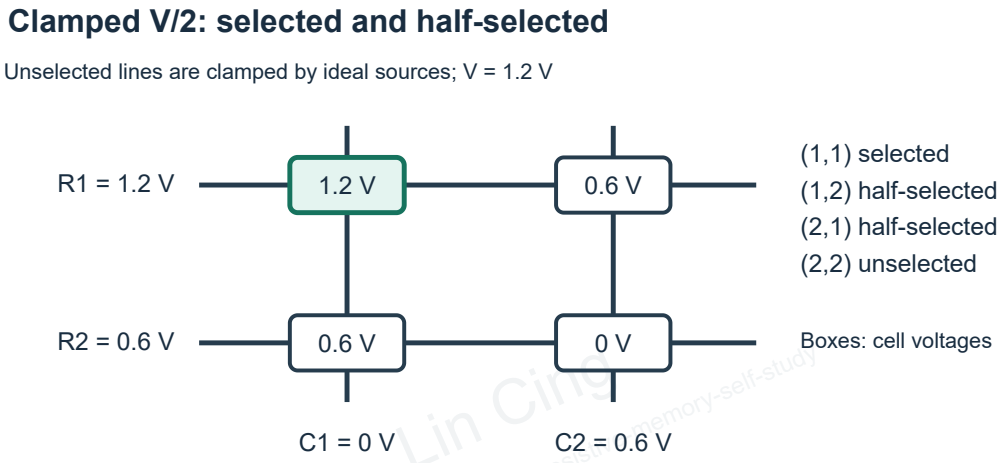
Drive the selected row to V, the selected column to 0 and every unselected row and column to $V/2$. Assume ideal voltage sources and wires.

Cell position	Row potential	Column potential	Row − column
Selected intersection	V	0	V
Selected row only	V	$V/2$	$V/2$
Selected column only	$V/2$	0	$V/2$
Neither selected	$V/2$	$V/2$	0

For $V = 1.2$ V the half-selected cells see 0.6 V. This is one bias scheme; finite line resistance or voltage division within a 1S1R cell changes the device-level voltages.

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Fixed-bias example: follow the sources



The unselected lines are **clamped**, not floating: their sources can supply or absorb current. The selected-column current can contain both the selected-cell current and current from a half-selected cell.

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Half-select is a stress condition

An unselected cell experiences a fraction of the applied pulse. Whether it remains unchanged depends on its initial state, device voltage, polarity, pulse width and number of repetitions—not on voltage alone.

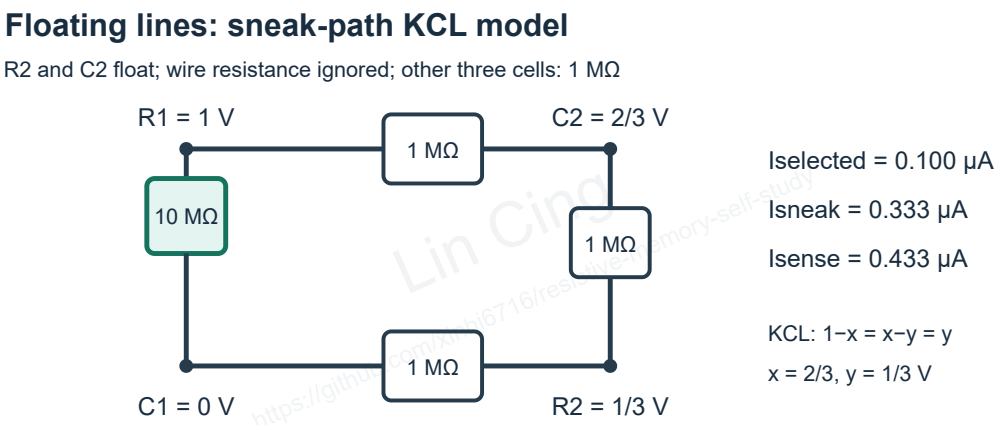
For an array, count the pulses imposed on each half-selected cell by its neighbors' writes. Compare its resulting distribution against the read criterion, including variability and cumulative disturb.

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Floating lines create a different network

Take a 2×2 passive array: row $R_1 = 1$ V and column $C_1 = 0$ V; R_2 and C_2 float. The selected cell R_1–C_1 has conductance $0.1\,\mu$S; each other cell has $1\,\mu$S.

Assume linear, fixed-state conductances, ideal wires and no connection from a floating line to an external load. A path $R_1 \to C_2 \to R_2 \to C_1$ now runs in parallel with the selected cell.



Solve the floating network by KCL

Let the unknown potentials of C_2 and R_2 be x and y volts. With all three unselected conductances equal to $1\,\mu$S, Kirchhoff's current law gives

$$(x-1)+(x-y)=0, \qquad (y-x)+y=0.$$

Thus $2x-y=1$ and $2y=x$, so $x=2/3$ V and $y=1/3$ V. Each of the three series branches drops $1/3$ V.

Selected-cell current is $0.1\,\mu$A; the alternative path carries $1/3\,\mu$A. The sense node at C_1 receives $0.1+1/3=0.433\,\mu$A, approximately.

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Compare the boundary conditions

In fixed $V/2$ bias the other lines are held by sources; in the floating example their voltages are **unknowns determined by KCL**. The three-resistor path calculation applies to that floating 2×2 topology, not automatically to a clamped array.

An HRS selected cell surrounded by LRS cells is an illustrative adverse pattern: the unwanted current can exceed the selected current. The actual worst pattern depends on the complete bias, device and sensing model.

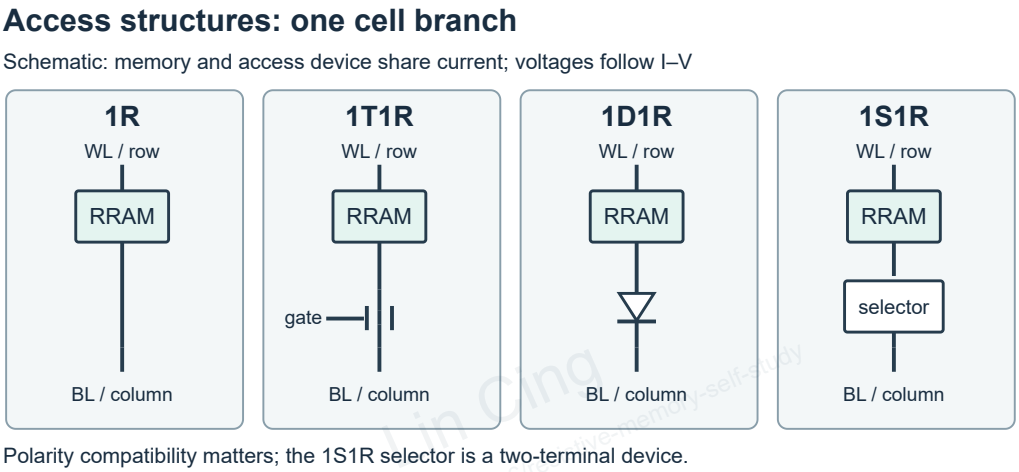
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Access structures change the available controls

Structure	Access mechanism	Design consequence
1R	Memory element alone	Isolation relies on its own I–V and bias scheme.
1T1R	A gated transistor in series	Gate control can suppress leakage and limit current, at a footprint and voltage cost.
1S1R	Two-terminal selector in series	Selector nonlinearity suppresses some low-bias currents without a gate.
1D1R	Diode in series	Rectification favors one direction; compatibility with both programming polarities must be checked.

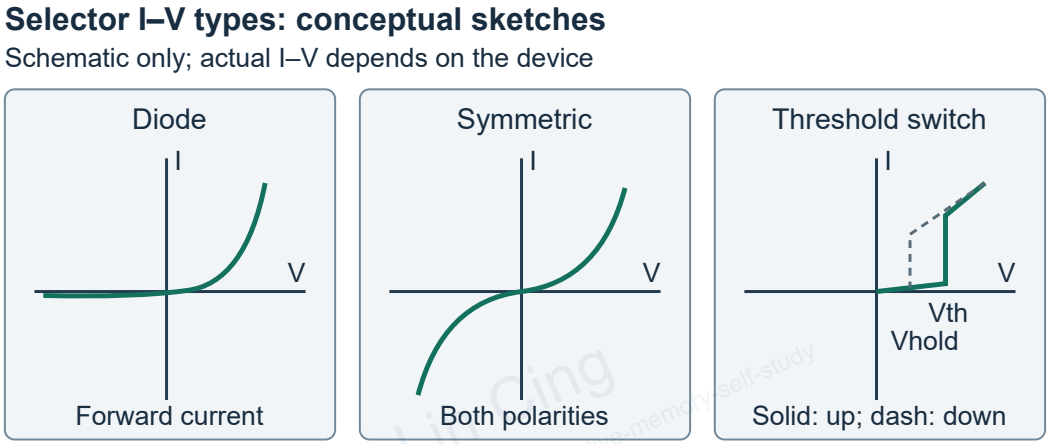
A self-selective element incorporates access nonlinearity into its stack; it does not provide the separate gate of 1T1R.

Access structures: follow each control path



The transistor has a separate gate terminal. A two-terminal selector or diode is driven through the series path; the memory and its access element share the applied voltage.

Three selector response families



A **rectifier** conducts asymmetrically in the two polarities. A **varistor** gives a strongly nonlinear response; it may be engineered for conduction in both directions, but symmetry is not automatic. A **threshold switch** can enter a conducting state after a threshold and return when its holding condition is lost.

Each label describes a response family, not a guarantee of leakage, switching speed, retention or suitability for a given memory.

Threshold and holding conditions

For a hypothetical volatile selector, an increasing voltage reaches a turn-on threshold V_{th}; after turn-on, the current and device voltage follow the selector's conducting branch. The selector returns to its off branch when current falls below a holding current I_h in the stated model.

The conducting branch can have hysteresis, and real thresholds depend on polarity, pulse duration and history. A threshold selector is not an ideal permanent latch and must not be confused with the nonvolatile state stored in RRAM.

The source, wire and series memory constrain the trajectory: crossing an isolated-device threshold alone does not prove a stable operating point.

A series selector shares voltage with memory

The same current flows through both devices and their voltages satisfy $V_{app} = V_S + V_M$. For nonlinear devices, solve $I_S(V_S) = I_M(V_M)$ together with that sum, using the relevant state, polarity and branch.

For a **fixed-state linear illustration**, let $R_S = 2\ \text{k}\Omega$, $R_M = 4\ \text{k}\Omega$ and $V_{app} = 1.2\ \text{V}$. Then $I = 1.2/6\,\text{k}\Omega = 0.2\ \text{mA}$; $V_S = 0.4\ \text{V}$ and $V_M = 0.8\ \text{V}$. Neither element receives the full applied voltage.

For a threshold selector use its actual I–V branches and circuit load line; fixed-resistance division cannot predict the switching transient or the holding condition.

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A nonlinear operating point: a piecewise model

Assume a positive-bias, non-hysteretic selector with $I = V_S/(100\,\text{k}\Omega)$ for $V_S < 0.8\ \text{V}$, and $I = (V_S - 0.8\ \text{V})/(1\,\text{k}\Omega) + 8\ \mu\text{A}$ for $V_S \geq 0.8\ \text{V}$. A fixed $4\,\text{k}\Omega$ memory is in series; $V_{app} = 1.2\ \text{V}$.

Try the low-voltage branch: $I = 1.2\ \text{V}/(104\,\text{k}\Omega) = 11.54\ \mu\text{A}$ and $V_S = 1.154\ \text{V}$. That violates $V_S < 0.8\ \text{V}$, so reject it.

On the high branch, $V_S = 0.8\ \text{V} + (1\,\text{k}\Omega)(I - 8\ \mu\text{A})$ and $1.2\ \text{V} = V_S + (4\,\text{k}\Omega)I$. The intersection is $I = 81.6\ \mu\text{A}$, $V_S = 0.8736\ \text{V}$ and $V_M = 0.3264\ \text{V}$, consistent with the branch. This is a teaching model, not a measured selector.

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Check the read and write windows separately

At the intended read voltage, distinguish HRS and LRS while limiting leakage and avoiding read disturb. At both SET and RESET polarities, the combined stack must deliver the memory's required voltage, current and pulse duration.

Also check selector on-current, half-select leakage, endurance, temperature, speed and variability. A selector that blocks reverse current may be suitable for a particular unipolar scheme but conflict with bipolar operation; topology alone cannot establish compatibility.

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What a nonlinearity ratio tells us

For a specified device state, polarity and measurement timing, define

$$NL = \frac{|I(V_{read})|}{|I(V_{read}/2)|}.$$

The denominator must be nonzero, and numerator and denominator must use the same current sign or magnitudes consistently. For $V_{read} = 1.2\,\mathrm{V}$, $NL = 10^3$ and $|I(1.2\,\mathrm{V})| = 10\,\mu\mathrm{A}$, the model gives $|I(0.6\,\mathrm{V})| = 10$ nA.

This synthetic example compares only two points. It does not specify other voltages, switching dynamics, noise or complete-array read margin.

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Read margin needs a sensing definition

Let I_H and I_L denote the sensed currents for a selected HRS and LRS, including all paths admitted by the chosen bias circuit. In a simple current-threshold reader, choose T and call the state LRS if $I_{sense} > T$.

If the possible HRS currents lie in $[H_{min}, H_{max}]$ and LRS in $[L_{min}, L_{max}]$, a deterministic threshold separating both ranges exists only if $H_{max} < L_{min}$. The separation $L_{min} - H_{max}$ is a current-domain margin under this definition.

Changing the sense circuit or using a voltage-domain margin changes the calculation; noise and distributions must be included before claiming an error rate.

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A small current-margin calculation

Use the previous floating 2×2 network, assume the three unselected cells remain linear at $1\,\mu$S, and switch only the selected conductance between 0.1 and $1\,\mu$S. Both states see 1 V across the selected cell and have the same $1/3\,\mu$A alternative-path current.

Hence $I_H = 0.433\,\mu$A and $I_L = 1.333\,\mu$A. With threshold $T = 0.8\,\mu$A, the HRS and LRS separations from threshold are 0.367 and $0.533\,\mu$A respectively. These nominal values assume an ideal current sense at 0 V, no noise and no state variation.

If unselected states or node voltages vary between reads, recalculate both ranges. The example does not establish how margin scales for a larger array.

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Threshold window with an assumed background

For a separate example, hold the sense column at 0 V and assume the sneak contribution is fixed at $0.3\,\mu$A regardless of the selected state. Let the selected HRS and LRS currents be 0.1 and $1.0\,\mu$A, respectively. The sensed currents are then $I_H = 0.4\,\mu$A and $I_L = 1.3\,\mu$A.

With the rule "LRS if current is strictly greater than T," both nominal states are classified correctly when $0.4 \le T < 1.3\,\mu$A. This definition gives a current window $I_L - I_H = 0.9\,\mu$A; other definitions may normalize the margin differently.

The $0.3\,\mu$A is an explicit simplifying assumption, not the $1/3\,\mu$A computed for the earlier floating network. In general solve the entire network separately for HRS and LRS.

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Add variability and bounded sensing error

In a particular operating condition, suppose measured HRS currents lie in $[0.35, 0.55]\,\mu$A and LRS currents in $[1.10, 1.40]\,\mu$A. Let the sensing error have absolute value at most $0.05\,\mu$A. The resulting observed ranges are $[0.30, 0.60]$ and $[1.05, 1.45]\,\mu$A.

Using the same strict-greater-than threshold rule, every value in these bounds is classified correctly for $0.60 \le T < 1.05\,\mu$A. If the ranges overlap, no single threshold guarantees correct classification for all bounded cases.

This is a bounded-error example. Actual distributions, sample sizes, correlations and failure rates need separate evaluation.

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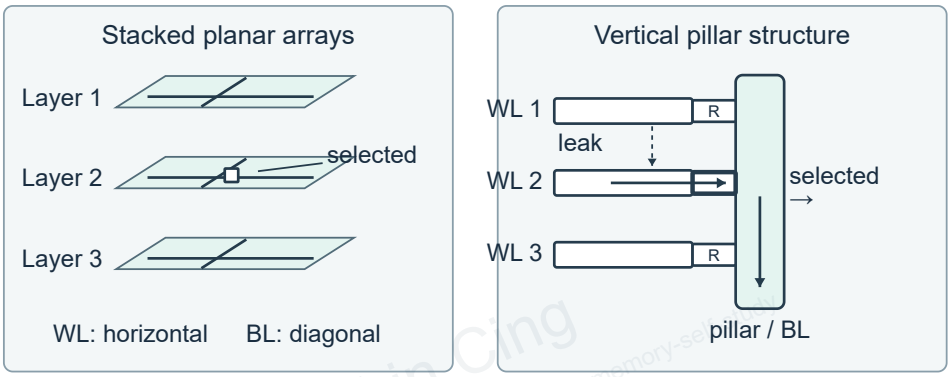
Array size: where leakage accumulates

Larger arrays generally introduce more possible paths and longer wires, but the actual sensed leakage depends on topology, biases and nonlinear I–V characteristics. A sum of identical half-select currents is valid only if the assumed circuit really presents those same device voltages and currents.

To evaluate a proposed size, solve the network for representative data patterns and process corners, then include sense noise, timing, supply and reference uncertainty. A single two-point NL cannot supply these missing conditions.

Horizontal layers and vertical pillars

3D integration: selected path and inter-layer leakage



Leakage depends on structure and bias.

Horizontal stacking repeats lateral crossbar planes. In a vertical structure, memory cells occur where a pillar electrode meets separately addressable stacked word-line layers. The selected path and isolation depend on the particular layout and line bias.

Interlayer leakage, alignment, parasitic capacitance, contacts, heat and yield can offset the apparent density gain. Different 3D architectures cannot be ranked without a stated circuit and manufacturing comparison.

Interpret an area calculation correctly

For an idealized planar crosspoint with pitch $2F$ in each direction, area per cell is $(2F)^2 = 4F^2$. If $F = 20$ nm, this is $1600\,\mathrm{nm}^2$.

With three perfect layers over the same projected footprint and one bit per cell, the ideal projected cell area per bit is $1600/3 \approx 533.3\,\mathrm{nm}^2$. This is an accounting estimate, not the area of a manufactured memory chip.

Peripheral decoders, sensing, routing, redundancy, layer isolation, spacing and yield must be included in a system comparison. Extra layers can change process cost and thermal constraints.

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From a nanoscale pattern to addressable cells

AAO: anodize aluminium to form a porous template, then deposit or grow material through pores. A regular local pattern does not itself register each device to external circuitry.

Nanosphere lithography: assemble a sphere monolayer and use it as a mask for deposition or etching. Particle placement, defects, removal and registration limit usable arrays.

Block-copolymer directed self-assembly (DSA): form periodic domains guided by a prepattern, then selectively remove or convert domains and transfer the pattern. Guidance, defect inspection, etch selectivity and alignment to contacts are part of the process.

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Pattern formation is not system integration

A useful fabrication chain is **pattern formation → pattern transfer → layer alignment → contacting and addressing → electrical and yield testing**. Nanoscale pitch is only the first constraint.

Check whether the switching layer remains continuous between cells: lateral conduction can create crosstalk. Compare pattern defects and placement error as well as feature size; a dense microscopic pattern is not automatically a functional high-yield memory array.

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A reproducible array comparison

State the stack, access structure, array dimensions, line biases, reference polarity, wire and sense models, pulses and timing. Record whether the reported current includes unselected paths and whether the result comes from measurements, simulation or extrapolation.

Report temperature, sample count, cycle history, variation, failure criterion and relevant uncertainty. A particular experimental selector or 3D stack is evidence for that configuration, not a universal specification for all RRAM arrays.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Redox mechanisms: from ion motion to memory

An oxide memory is a mixed ionic–electronic system. Its electrical response depends on mobile defects, electrode reactions, thermal history and measurement time. This unit develops the conditions needed to connect an observed I–V curve to a physical mechanism.

All numerical examples are hypothetical teaching models; they are not device specifications or reported measurements.

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Separate three observations

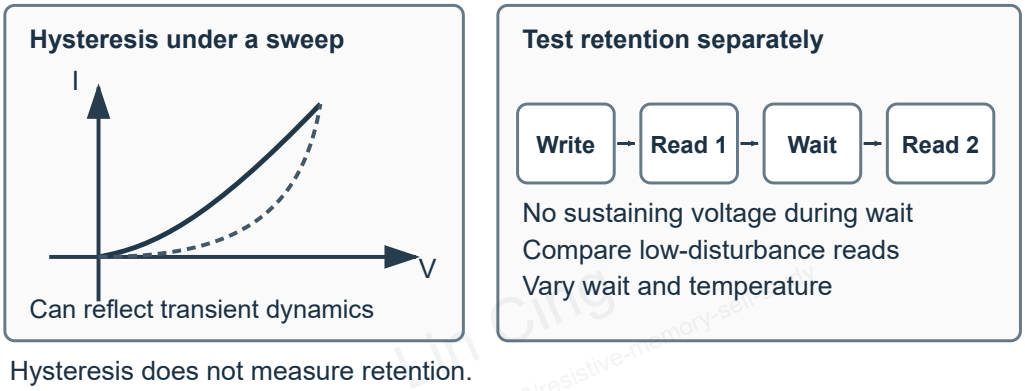
Switching is a change in conductance caused by an operation. **Retention** asks whether a distinguishable state remains after the driving signal is removed. **Short-term hysteresis** means the response depends on recent history over the measurement time.

A loop in a voltage sweep can occur when ion and electron responses have different timescales. It does not, by itself, establish long-term nonvolatile memory. Conversely, two retained states can be distinguished using a suitably non-disturbing read protocol without relying on the shape of a continuous sweep.

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A measurement distinguishes timescales

I–V hysteresis alone does not prove nonvolatile memory



Hysteresis does not measure retention.

The sketch compares two protocols; it does not assert a specific material or quantitative retention time. The waiting period must be specified and the read pulse checked for disturbance.

From material change to terminal current

One possible chain is: an applied pulse changes an electrode reaction and/or moves ionic defects; a nonuniform composition develops; that distribution modifies an electronic conduction path or interface barrier; the read current changes.

These links are hypotheses to be tested. An I–V loop alone cannot tell whether conduction is filamentary, interface-controlled, or distributed across the film. A mechanism should also predict spatial, chemical and time-dependent observations.

Charge carriers and their roles

The electronic current can be dominated by electrons or holes, while much slower ionic defects alter the device structure. The measured terminal current can include electronic, ionic and displacement contributions:

$$I_{\mathrm{total}} = I_e + I_{\mathrm{ion}} + I_{\mathrm{disp}}.$$

In a one-dimensional series cross section, charge continuity gives the same total current at each position; individual contributions can vary with position and time as charge accumulates. An ammeter does not automatically measure ion flow.

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Reaction bookkeeping: an oxygen vacancy

For oxygen leaving an oxide into an oxygen reservoir, one possible idealized reaction is

$$O_O^{\times} \rightleftharpoons \tfrac{1}{2}O_2 + V_O^{\bullet\bullet} + 2e'.$$

Here $O_O^{\times}$ is oxygen on its lattice site, $V_O^{\bullet\bullet}$ is a vacancy with effective charge +2 relative to the lattice, and e' denotes an electron with effective charge −1. Both atom count and effective charge balance. This is a model of one charge state, not a claim that all vacancies have this charge.

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When may vacancy concentration predict electrons?

If vacancies are the only significant positively charged defects, electrons the only significant negative carriers, and local charge neutrality applies, then $n \simeq 2[V_O^{\bullet\bullet}]$.

For $[V_O^{\bullet\bullet}] = 2 \times 10^{18}\,\mathrm{cm}^{-3}$, the model gives $n \simeq 4 \times 10^{18}\,\mathrm{cm}^{-3}$. With negatively charged acceptors included, neutrality becomes $2[V_O^{\bullet\bullet}] = n + N_A^-$: for $[V_O^{\bullet\bullet}] = 3 \times 10^{18}$ and $N_A^- = 2 \times 10^{18}\,\mathrm{cm}^{-3}$, $n = 4 \times 10^{18}\,\mathrm{cm}^{-3}$ instead of $6 \times 10^{18}\,\mathrm{cm}^{-3}$ when acceptors are incorrectly omitted. Interface space charge can also invalidate local neutrality.

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ECM: metal transfer and reduction

In an electrochemical metallization (ECM) cell with an active Ag electrode, a possible oxidation reaction is $\mathrm{Ag} \rightarrow \mathrm{Ag}^+ + e^-$. Ag ions enter the switching layer; reduction $\mathrm{Ag}^+ + e^- \rightarrow \mathrm{Ag}$ can deposit metal elsewhere.

Where growth begins and how a path forms depend on ion transport, electron supply, interfaces and rates. The equations do not imply that every ECM cell has the same filament-growth direction or that SET has one universal polarity.

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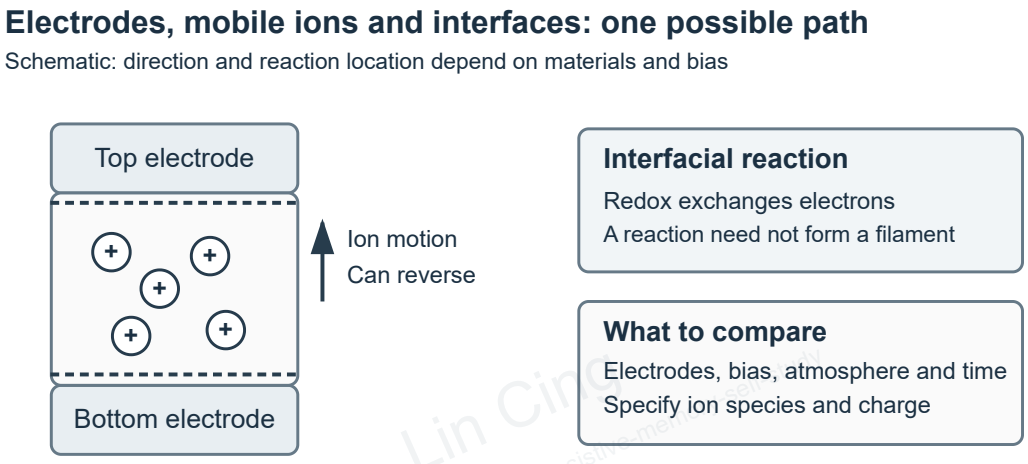
VCM: oxygen exchange and vacancy redistribution

In a valence-change mechanism (VCM), oxygen chemical potential, vacancy populations and local transition-metal valence can change. A reduced path, a narrow gap near an electrode, or a modified interface region can alter electronic transport.

VCM need not mean a metallic filament; ECM need not make every observed filament metallic throughout its lifetime. Composition-sensitive and spatially resolved measurements are needed to distinguish models.

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Follow ions and interfaces in a cell



The illustration shows a possible direction of ion motion, not an assigned electrode polarity or a specific reaction. For an Ag-containing ECM stack, the reactions on the preceding page are examples; actual growth direction depends on the stack, voltage and kinetics.

Electrochemical potential and driving force

For species i with charge number z_i, electric potential ϕ and chemical potential μ_i, the electrochemical potential is

$$\widetilde{\mu}_i = \mu_i + z_i e \phi.$$

At fixed temperature, a gradient of $\widetilde{\mu}_i$ drives transport. A concentration gradient contributes through μ_i; the electric field $E = -\partial\phi/\partial x$ contributes through $z_i e\phi$. Electric field alone does not determine net transport when concentration gradients are present.

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Drift and diffusion in a limited model

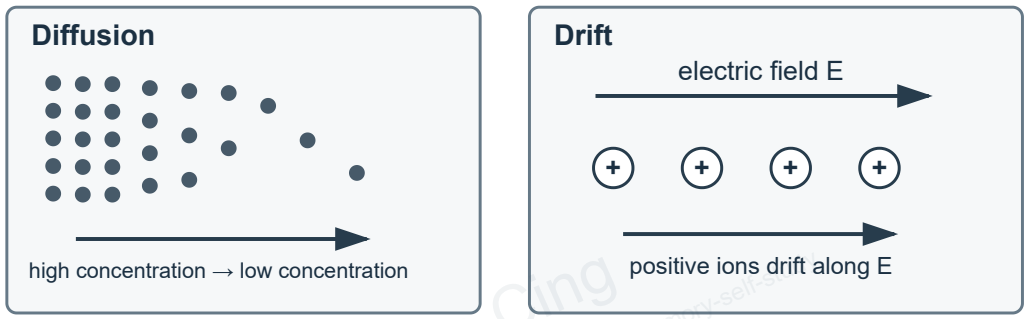
For dilute, noninteracting particles with concentration $c(x)$, constant D, charge number z, and the Einstein relation, the one-dimensional number flux is

$$J = -D\frac{\partial c}{\partial x} + \frac{Dze}{k_B T} cE.$$

Here J is number per area per time, not electrical current density; the latter is $j_{\mathrm{ion}} = zeJ$. For concentrated defects, strong fields, interactions or nonuniform temperature, use an appropriate nonideal transport model rather than this expression unchanged.

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Compare diffusion with electric drift



Net flux combines both terms and boundary conditions; electron flow is separate.

The arrows are for the stated positive defect and illustrated gradients only. Changing charge sign or either gradient can reverse an individual contribution; the net flux is their vector sum.

Drift and diffusion: calculate both terms

For the dilute model, let $D = 10^{-18}\,\mathrm{m^2/s}$, $z = 2$, $E = 10^{8}\,\mathrm{V/m}$, $c = 10^{24}\,\mathrm{m^{-3}}$, $dc/dx = 10^{30}\,\mathrm{m^{-4}}$ and $k_B T = 4.14 \times 10^{-21}$ J.

The diffusion flux is $-10^{12}\,\mathrm{m^{-2}s^{-1}}$. With $e = 1.60 \times 10^{-19}$ C the drift contribution is $+7.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$; at this precision the net flux is also approximately $+7.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$. These are particle fluxes, not amperes.

A field does not erase a migration barrier

Suppose a 12 nm film has a uniform 0.6 V drop. The electric-field magnitude is 0.05 V/nm. Across a 0.2 nm hop along the electric field the potential falls by 0.01 V; for a hypothetical $z = +2$ defect the endpoint electrostatic energy changes by -0.02 eV.

This compares endpoint energies. The reduction of the actual activation barrier depends on the transition-state position and energy landscape and is not automatically 0.02 eV. A highly nonuniform drop at an interface changes the estimate.

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Interface reaction versus bulk transport

An electrode can exchange oxygen with the oxide, store oxygen, react to form an interlayer, or approximately block exchange. A blocking boundary constrains normal flux there; it does not imply that the defect concentration is uniform inside the film.

If supply at one interface is slow compared with redistribution in the oxide, defects can accumulate in a different region from the fast-supply case. Growth direction and the active gap therefore depend on **both** boundary reaction kinetics and bulk transport, not electrode name alone.

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A supply–transport balance, with limits

For one illustrative slab of thickness L and cross-sectional area A, let $N(t)$ be its spatially averaged defect concentration, with injection flux s and removal flux r on opposite boundaries. With no net volumetric reaction, conservation gives $d(ANL)/dt = A(s - r)$, hence

$$\frac{dN}{dt} = \frac{s - r}{L}.$$

Fluxes may depend on concentration, voltage, temperature and time. This averaged balance cannot determine a local profile, filament geometry, or which side the filament grows from; those require the transport equation and boundary conditions.

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Crossing an assumed concentration threshold

Take the simplified constant-flux model with $L = 10\ \text{nm} = 10^{-6}\ \text{cm}$, $s = 8 \times 10^{12}$ and $r = 2 \times 10^{12}\ \text{cm}^{-2}\text{s}^{-1}$. Then $dN/dt = 6 \times 10^{18}\ \text{cm}^{-3}\text{s}^{-1}$.

If $N(0) = 2 \times 10^{18}\ \text{cm}^{-3}$ and a specified model threshold is $N_c = 5 \times 10^{18}\ \text{cm}^{-3}$, then $N(t) = N(0) + (s - r)t/L$ reaches N_c after $(5 - 2)/6 = 0.5$ s. This is an average crossing, not proof of a local metal transition; a real threshold depends on material and spatial distribution.

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A separate steady-state spatial model

Now consider a different one-dimensional model: constant D, no drift, no volume sources, and fixed endpoint concentrations. Steady conservation gives $dJ/dx = 0$, and Fick's law gives $J = -D\,dc/dx$. Thus $c(x)$ is linear and the flux is uniform.

Take $L = 10\ \text{nm} = 10^{-6}$ cm, $D = 10^{-14}\,\text{cm}^2/\text{s}$, $c(0) = 8 \times 10^{18}$ and $c(L) = 2 \times 10^{18}\,\text{cm}^{-3}$. Then

$$c(x) = \left(8 - 6\frac{x}{L}\right) 10^{18}\,\text{cm}^{-3}, \quad J = +6 \times 10^{10}\,\text{cm}^{-2}\text{s}^{-1}.$$

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Where does the local threshold hold?

Using that steady diffusion solution and the illustrative threshold $N_c = 5 \times 10^{18}\,\text{cm}^{-3}$, solve $8 - 6x/L \geq 5$: the interval is $0 \leq x \leq L/2$, or 0–5 nm, of length 5 nm.

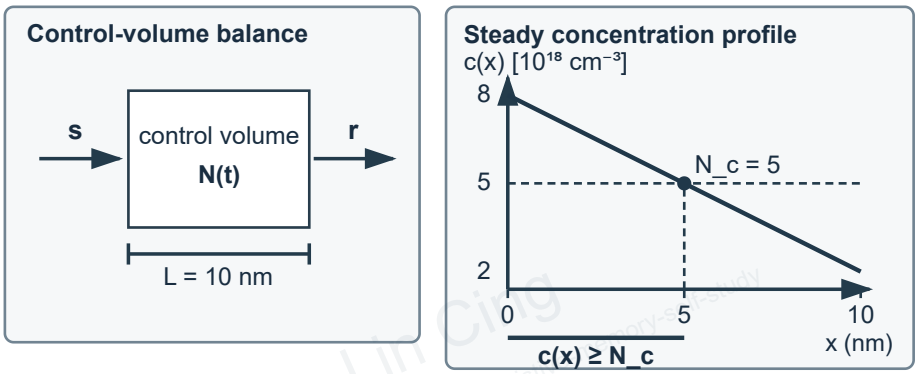
The assumed boundary concentrations and steady-state profile differ from the earlier transient control-volume example; do not combine the two as one time history. Crossing a chosen concentration threshold in this 1D model is not a general prediction of a connected, metallic filament.

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Boundary flux and local profile

Control-volume balance and a concentration profile

Teaching examples: the two models are separate



A threshold alone proves neither phase change nor filament.

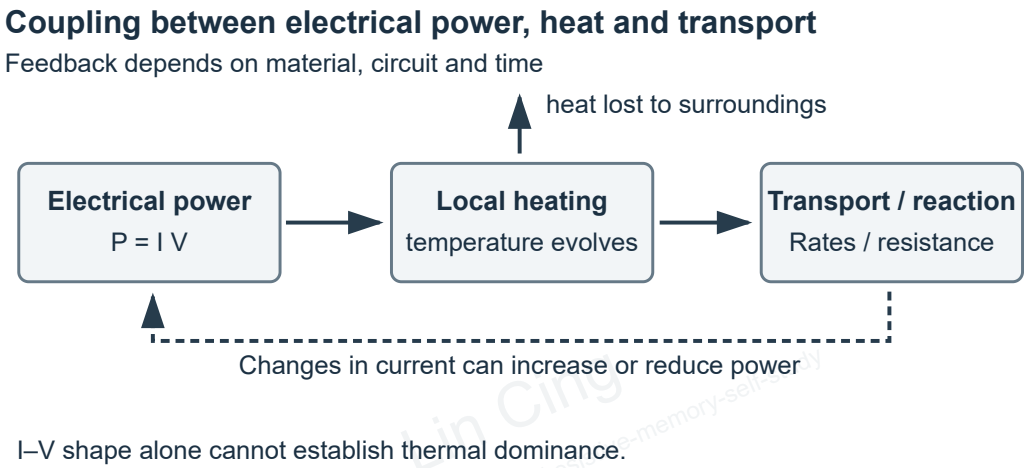
The left panel is a transient average; the right panel is a separate fixed-boundary steady state. Their fluxes and concentrations are not one continuous experimental sequence.

Forming, compliance and feedback

Some devices require electroforming: an initial electrical treatment that establishes a switchable state. Field-driven reactions and local Joule heating can reinforce one another; the evolving structure changes the local field and current.

Compliance limits the terminal current under the stated circuit and timing. Finite response time and parasitic capacitance can still produce a transient overshoot, so the set point alone does not fix local peak current, temperature or filament dimensions. Other devices are forming-free under their specified preparation and test conditions.

Feedback can amplify a local change



I–V shape alone cannot establish thermal dominance.

This loop is a possible mechanism under a specified circuit: increasing temperature need not increase conductance in every material or bias regime. Compliance and thermal loss limit the process differently.

Why a nanometre gap matters

Take a deliberately simple model in which read current through a barrier is $I(g) = I_0 e^{-2\kappa g}$ with fixed I_0 and $\kappa = 5\,\mathrm{nm}^{-1}$. A gap change from $g_1 = 1.0$ nm to $g_2 = 1.3$ nm gives

$$\frac{I(g_2)}{I(g_1)} = e^{-2(5)(0.3)} = e^{-3} \simeq 0.0498.$$

Thus the wider-gap current is about 5% of the initial current. This is an exponential sensitivity illustration, not a complete tunnelling theory: barrier height, bias, geometry and prefactor may also change.

Interface control is not proof of a filament

An interface barrier can limit the current even when the remainder of the oxide is relatively conductive. Changes in charge distribution and local doping can modify barrier width, band bending and available transport channels.

At fixed temperature and device geometry, ask whether an area series, local current map, spectroscopy and cross-sectional imaging support a distributed interface or localized path. No single area-scaling result proves the entire current topology without checking confounders.

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Bipolar and unipolar are operating descriptions

In **bipolar** operation, the two transitions are induced under opposite voltage polarities as defined for that device. In **unipolar** operation, both can occur at the same polarity under different pulse, current, thermal or circuit conditions.

The label classifies a protocol, not a unique microscopic mechanism. In particular, same-polarity RESET does not prove a particular temperature or diffusion process without supporting measurement.

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Compare the energy of voltage scans

For a voltage ramp beginning at zero with constant positive scan rate $v = dV/dt$, substitute $dt = dV/v$ into $W = \int V I \, dt$ to obtain

$$W = \frac{1}{v} \int_0^{V_{\mathrm{end}}} V I(V) \, dV.$$

Use the current actually measured along each scan, including its state history; this change of variables does not assume SET and RESET share the same I–V curve. Different ramp rates and switching points must be included explicitly.

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A same-polarity scan comparison

Take two hypothetical positive-going scans, each at $v = 1$ V/s. For the SET scan let $I = aV$ with $a = 1\,\mu$S up to 1 V; for RESET use $a = 4\,\mu$S up to 0.5 V. Integrating $VI = aV^2$ gives

$$W_{\mathrm{SET}} = \frac{1\,\mu\mathrm{S}}{1\,\mathrm{V/s}} \frac{1^3}{3} = \tfrac{1}{3}\,\mu\mathrm{J}, \quad W_{\mathrm{RESET}} = \frac{4\,\mu\mathrm{S}}{1\,\mathrm{V/s}} \frac{0.5^3}{3} = \tfrac{1}{6}\,\mu\mathrm{J}.$$

Higher current at a given voltage alone does not determine integrated energy when voltage ranges and durations differ. Even if one scan delivers more energy, that alone cannot determine which reaches a higher peak temperature: the heat-loss path and time dependence also matter.

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Power, energy and temperature are different

Instantaneous electrical power is $P(t) = V(t)I(t)$ and delivered energy is $W = \int P(t)dt$. The same energy can produce different peak temperatures if pulse widths, heat loss and thermal paths differ.

For a lumped model, $C_{\mathrm{th}}\, d\Delta T/dt = P(t) - \Delta T/R_{\mathrm{th}}$. With constant power, $\Delta T(t) = PR_{\mathrm{th}}(1 - e^{-t/(R_{\mathrm{th}}C_{\mathrm{th}})})$. This model is not a spatial temperature map.

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A thermal pulse worked through

Assume $V = 1$ V and $I = 100\,\mu$A remain constant, so $P = 0.1$ mW. Let $R_{\mathrm{th}} = 10^6$ K/W, $\tau_{\mathrm{th}} = 100$ ns, and initially $\Delta T = 0$. The implied $C_{\mathrm{th}} = \tau_{\mathrm{th}}/R_{\mathrm{th}} = 10^{-13}$ J/K; the steady rise is $PR_{\mathrm{th}} = 100$ K.

After 100 ns the idealized rise is $100(1 - e^{-1}) \simeq 63.2$ K. Changing the thermal resistance, heat capacity or pulse shape changes the result; measuring electrical energy alone does not reveal peak local temperature.

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Retention and activation: a toy estimate

An activated process is often approximated by $\tau = \tau_0 \exp(E_a / k_B T)$ when one rate-limiting barrier dominates and temperature is constant. For $\tau_0 = 10^{-12}$ s, $E_a = 1.0$ eV and $k_B T = 0.02585$ eV at 300 K,

$$\tau \simeq 10^{-12} e^{38.68} \simeq 6.3 \times 10^{4}\ \mathrm{s} \approx 17.5\ \mathrm{h}.$$

This hypothetical time is not a retention prediction for a particular device. Real retention depends on barrier distributions, geometry, read criterion and other mechanisms; the fitted prefactor must be stated.

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Read–wait–read: test memory directly

Program two populations to HRS and LRS, remove the programming bias, wait at controlled temperature, then apply the same low-disturb read to each. Repeat for multiple times and devices; report read voltage, pulse duration, temperature and the state distributions.

Retention is a criterion on the separation and error rate of those distributions, not merely a single resistance value. Check that repeated reads do not themselves change the state; include drift and uncertainty.

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What an I–V crossing can and cannot show

In a simplified passive, non-battery-like device model, negligible displacement and ionic currents at low frequency can make the terminal curve pass near the origin. Even if a curve passes exactly through the origin, that is **not sufficient** to establish retention; an internal state may relax.

Offsets from capacitance, electrochemical electromotive forces, instrumentation and finite sweep rate can shift the crossing. These assumptions must be checked before interpreting a crossing. Compare sweep rates and perform the read–wait–read experiment.

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Build a falsifiable mechanism

1. Specify electrode stack, polarity convention, forming and compliance circuit, pulse timing, atmosphere and temperature.
2. Compare hypotheses using predictions for spatial current, element distribution, state evolution and temperature—not only a fit to one I–V curve.
3. Test alternative explanations and controls, including interface changes, parasitic current, measurement history and sample-to-sample variation.

An original model is useful when it explains the data within its assumptions and can be challenged by another experiment.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Seeing a switching device: measurements and evidence

In a nanoscale device, electrical response is a measured outcome; a filament, its composition and its motion are explanations that require additional evidence. This chapter connects an experiment, its signal and the limits of the conclusion.

Numerical examples are hypothetical. Images are original schematics; experimental figures and published performance values are not reproduced.

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Start with an evidence chain

Ask four questions in order: **What was driven? What was measured? How was it calibrated? Which mechanisms remain compatible with the data?**

An I–V change shows an electrical state change under the stated protocol. A time-correlated image supplies structural evidence; chemical analysis and control measurements can narrow the interpretation. No single contrast feature automatically establishes composition, valence, or a unique atomic mechanism.

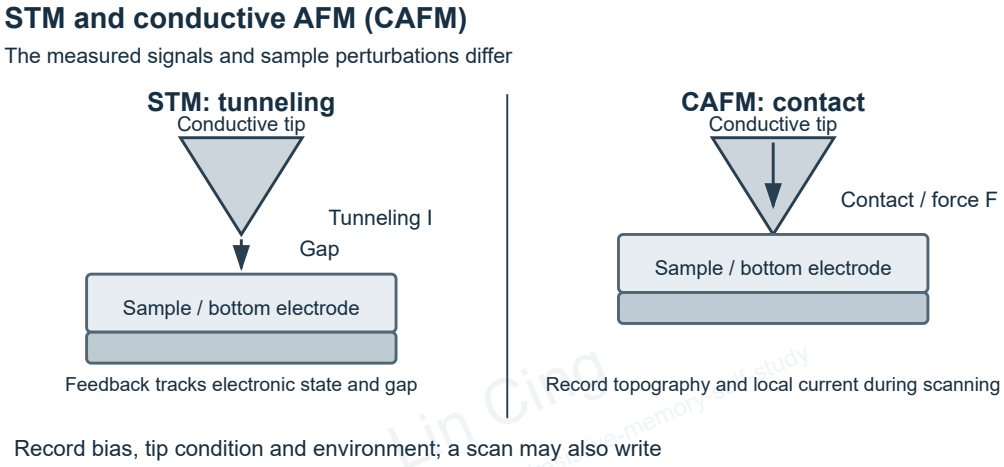
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Choose the probe for the question

Method	Primary signal	Useful question	Main limitation
in situ TEM	Transmitted-electron image, optionally EDS/EELS	How does a visible structure evolve under bias?	Thin specimen, beam and vacuum may perturb it.
STM/STS	Tunneling current or differential conductance	How does a local surface electronic response change?	Surface, tip and feedback influence the signal.
Conductive AFM	Contact current and topography	Where does current enter an accessible surface?	Contact and scan can alter the sample.
Scalpel SPM / PMCM	Sequential contact maps / force-modulated electrical response	What depth structure or mechanically sensitive region is inferred?	Reconstruction or pressure introduces additional assumptions.

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Link the device and the instrument



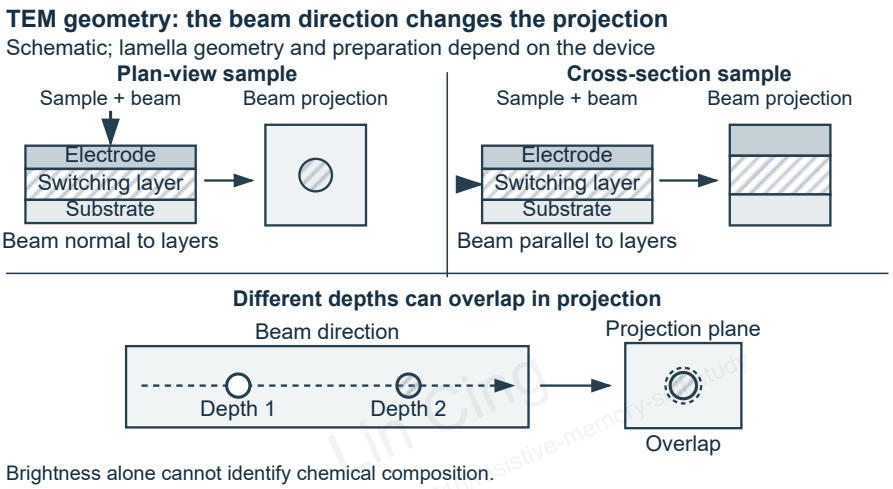
Record geometry, electrode assignments, voltage polarity, compliance, timing, environment and prehistory. The probe's electrical boundary is part of the experiment: a tip electrode is not automatically equivalent to an intact device's top electrode.

In situ TEM: what is actually observed?

An electron beam passes through a sufficiently thin specimen. The image is a **projection** of electron interactions through its thickness, not a direct three-dimensional concentration map. Biasing the specimen while acquiring images allows the electrical trace and structural evolution to be compared in time.

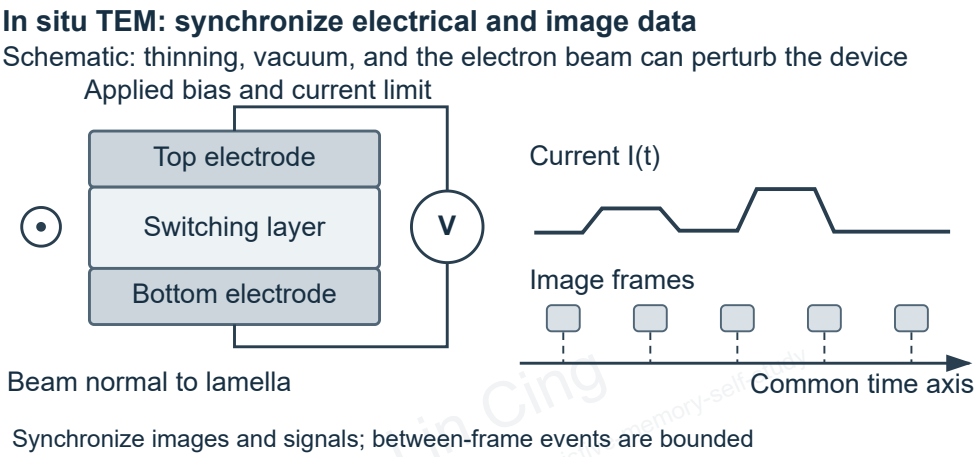
Different growth morphologies and nucleation positions are possible in different materials. A particular device's polarity or shape does not fix the direction for every ECM cell.

Cross-section and plan view: projections can overlap



In a cross-section view, objects separated along the beam can overlap as one feature in projection. For a deliberately ideal linear projection, two contributions of 2 and 3 arbitrary signal units sum to 5; that observed value alone cannot identify how many objects produced it. Real TEM contrast may be nonlinear and is not generally an additive concentration measurement. A plan-view projection looks from another orientation but still integrates along its own beam direction; tilting and matched electrical/chemical evidence help resolve the ambiguity.

Synchronize image and electrical trace



Synchronize images and signals; between-frame events are bounded

Before interpreting a feature, document which event is aligned with each frame, the exposure and clock uncertainty, and whether the device still switches after thinning. A frame is not necessarily an instantaneous sample: an exposure can average changes occurring during its interval.

A time-bound example

Suppose ideal instantaneous images are acquired every 30 ms. Frame 1 occurs at $t = 0$; frame 3 at 60 ms has no visible feature and frame 4 at 90 ms has one.

Under the ideal assumption that the feature becomes detectable once and remains detectable, its first detectable appearance is in $(60, 90]$ ms. The sequence alone does **not** give its exact onset, a speed or a chemical identity. Finite exposure, synchronization error and detection limits widen or complicate this inference.

Contrast, composition and oxidation state

Image contrast depends on specimen geometry, thickness, crystal orientation and imaging conditions as well as composition. A bright or dark region is evidence of contrast, not by itself a measurement of Ag, Cu or oxygen-vacancy concentration.

EDS provides elemental spectral information; EELS can provide elemental and chemical-state information when energy resolution, background and reference spectra support it. Both average over a finite interaction volume and can change the dose applied to the sample.

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TEM controls make a mechanism more credible

Compare an unbiased or lower-dose control, multiple switching cycles, and electrical behavior before and after imaging. Examine whether a morphology change coincides with a state change and whether composition-sensitive measurements support the proposed material.

FIB preparation, exposed surfaces, vacuum and the beam can affect oxygen content, interfaces, heating or damage. A non-switching thinned specimen cannot be silently treated as an undisturbed working device.

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Apparent motion has more than one explanation

Consider a cluster seen at position A in one frame and position B in a later frame. Translation is one hypothesis; dissolution near A and nucleation near B is another. Motion between frames or below resolution is not directly observed.

Vary time resolution and operating conditions, compare repeated trajectories and seek chemical or structural evidence. Report a mechanism as a supported model, not as the only possible explanation, when alternatives remain.

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STM: current across a tunneling gap

A conducting tip approaches a surface without the normal mechanical contact used in contact-mode AFM. At a fixed bias and small gap, the tunneling signal is strongly gap-dependent. Local surface states, tip states, the bias convention and the environment also influence current.

A pristine insulating film may not provide a stable measurable tunneling path; interpreting an image requires a specified sample stack and accessible surface.

In a simplified fixed-barrier model, $I \propto e^{-2\kappa z}$. If $\kappa = 10\,\mathrm{nm}^{-1}$ and the gap increases by 0.10 nm with all other parameters held fixed, $I_2/I_1 = e^{-2} \approx 0.135$. This is a model example, not a universal STM calibration.

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Feedback changes the meaning of an image

In **constant-current STM**, a feedback loop adjusts tip height to maintain a current setpoint while scanning. The recorded height reflects both geometry and local electronic response; a higher trace is not necessarily a physical protrusion.

In spectroscopy, an I–V trace and dI/dV depend on position, tip, bias, temperature and measurement protocol. Differential conductance may inform electronic states under a stated model; it is not an uncalibrated elemental map.

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A local write-and-read control

1. Image a region at a read condition chosen to minimize disturbance, and choose an unprogrammed control region.
2. Apply a specified local write pulse; record tip, pulse polarity, amplitude and duration.
3. Repeat low-disturb read at the same positions, then read again after a specified wait.
4. Compare current, topography and control drift. A persistent difference supports a retained local change under this protocol; it does not alone identify its chemical origin.

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Conductive AFM: contact current and topography

CAFM uses a conducting tip in mechanical contact with an accessible surface. A bias applied between the tip and another electrode yields a local current while the feedback system records topography. Tip force, coating, contact area, humidity, bias and current limit all affect the result.

The current map shows the response of **tip + contact + exposed sample + external circuit**, not an isolated, geometry-free property of the oxide.

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Two CAFM experiments with different boundaries

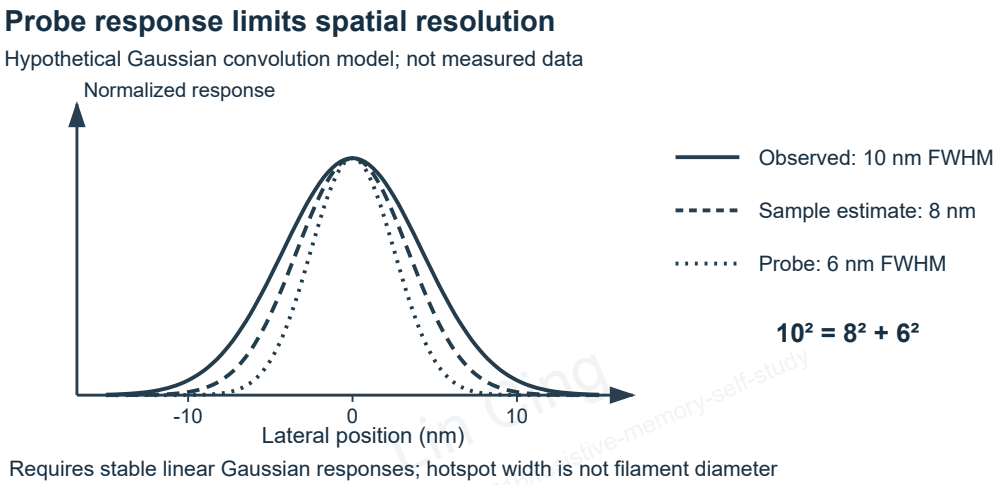
Tip-as-electrode: write and read directly on exposed film. The contact's field and force are part of the writing process.

Program first, expose later: switch a complete device electrically, remove its top electrode and map the exposed layer. Removal can damage or oxidize the feature, and the device cannot necessarily be cycled in the original configuration afterward.

Do not pool the two as if they had identical contacts, fields and sample histories.

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A current hotspot is not a direct diameter measurement



Requires stable linear Gaussian responses; hotspot width is not filament diameter

Measured width depends on the true spatial response, probe/contact geometry, sampling, feedback and noise. A visually wider hotspot can result from either a wider conducting region or a broader instrument response. Fit a stated forward model and show its limitations before inferring a size.

Gaussian resolution model: work through the numbers

Assume the true one-dimensional signal and a linear, shift-invariant point-spread function are Gaussian, with no additional broadening. Gaussian variances add; because $\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma$,

$$w_{\mathrm{obs}}^2 = w_{\mathrm{sample}}^2 + w_{\mathrm{tip}}^2.$$

For $w_{\mathrm{obs}} = 10$ nm and $w_{\mathrm{tip}} = 6$ nm, $w_{\mathrm{sample}} = \sqrt{10^2 - 6^2} = 8$ nm. The result is a model-dependent effective width, **not** a universal CAFM diameter correction.

If the two FWHMs have independent standard uncertainties of 0.5 nm, first-order uncertainty propagation gives $\sigma_w = \sqrt{(10 \cdot 0.5)^2 + (6 \cdot 0.5)^2}/8 \approx 0.73$ nm. This does not include uncertainty from model mismatch.

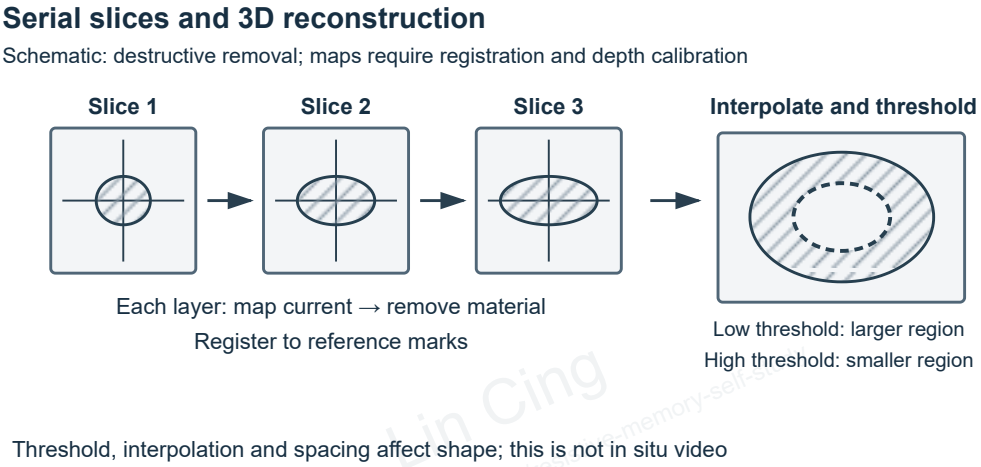
When deconvolution is not justified

If the measured width is no greater than the independently calibrated instrumental width, this Gaussian subtraction yields no positive real width. Within measurement uncertainty the feature may be unresolved; report an upper bound or a measurement limit rather than a fabricated diameter.

Even when subtraction is real, nonlinear contact, changing tip shape, non-Gaussian profiles and unknown uncertainty invalidate or weaken the estimate. Inspect raw profiles and calibration data.

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Scalpel SPM: a sequence, not a live 3D view



Program and locate a structure, remove material in controlled increments, and acquire a current map at each exposed surface. Register slices to a common reference and calibrate depth. The measurement is destructive and sequential; the sample may evolve between slices.

Reconstruction introduces choices

Spatial registration, depth estimates, interpolation, smoothing and the threshold used to draw an isosurface all affect the displayed shape. A continuous rendering is a visualization of measured slices **plus assumptions**, not independently measured data at every voxel.

Vary registration and threshold, use a stable reference electrode or separately characterized reference region at each slice to monitor tip changes, and report the slice spacing and removal uncertainty. A changing reference response flags possible tip wear or contamination; it cannot by itself rule out all local contact changes. Samples may oxidize or be mechanically damaged during exposure.

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Interpolate explicitly

At one registered lateral position, suppose measured currents are 2, 6 and 10 nA at depths $z = 0, 2, 4$ nm. With linear interpolation between the first two measurements,

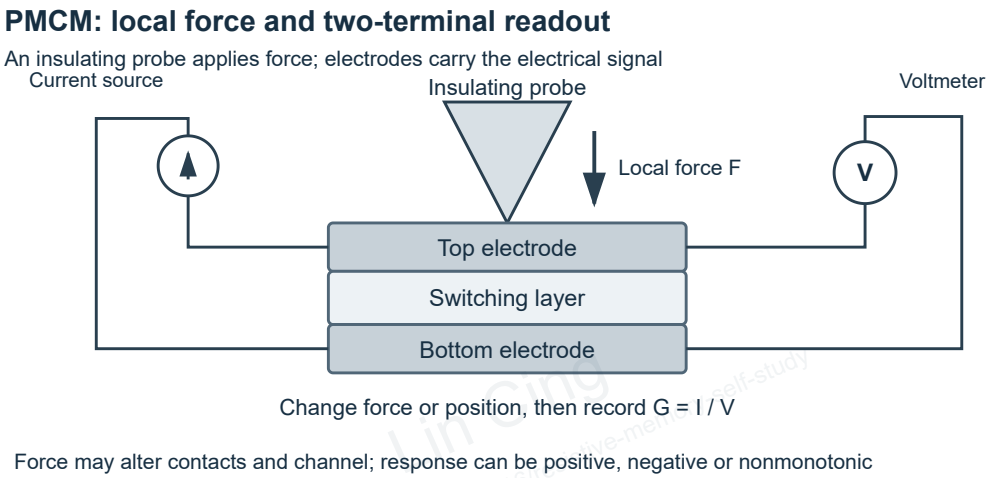
$$I(1\,\mathrm{nm}) = 2 + \frac{1-0}{2-0}(6-2) = 4\,\mathrm{nA}.$$

This 4 nA value is a model estimate, not a fourth measurement. An isosurface threshold of 3 nA includes different locations from one at 7 nA; a changing displayed volume is not by itself proof that the physical channel changed.

Similarly, interpolation between $z = 2$ and 4 nm gives $I(3) = 8$ nA and places the 7 nA crossing at $z = 2.5$ nm; both depend on the linear assumption.

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PMCM: a mechanical perturbation, an electrical readout



A non-conducting tip applies localized mechanical force to the top electrode of a complete device while the voltage across it is monitored under a low fixed current. Scanning the tip can reveal positions at which the whole-device resistance responds to that perturbation.

Force magnitude, load distribution, electrode mechanics, electrical bias, tip wear and spatial registration must be reported. The readout is not simply the current through the tip.

Calculate the conductance response

At fixed $I = 1\,\mu$A, suppose conductance rises from $G_0 = 10\,\mu$S to $G_1 = 12\,\mu$S when pressure is applied. Then

$$R_0 = 100\,\mathrm{k\Omega}, \quad R_1 = 83.33\,\mathrm{k\Omega}, \quad V_0 = 0.100\,\mathrm{V}, \quad V_1 \approx 0.0833\,\mathrm{V}.$$

The fractional conductance change is $(G_1 - G_0)/G_0 = 0.20$, or $+20\%$. A voltage dip at fixed current corresponds to lower resistance in this example; response sign and spatial shape depend on the device and mechanism.

What a PMCM contrast does and does not show

A repeatable force-correlated response identifies a mechanically sensitive region under specified conditions. It can guide cross-sectional observation, but does not by itself establish that every current path is there, reveal a unique filament geometry, or prove that conductance must rise under pressure.

Tunneling distances, electronic structure, mechanical deformation and quantum interference can contribute differently. Compare reverse scans, force-dependence and controls with other imaging and electrical measurements.

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Coherent transport: why the response may change sign

For a phase-coherent conductor at zero temperature and infinitesimal bias, the Landauer expression is $G = G_Q \sum_n T_n(E_F)$, where $G_Q = 2e^2/h$ assumes spin degeneracy and T_n is the transmission probability of mode n. At finite temperature the energy-dependent transmission is weighted over the thermal window; finite bias requires an appropriate non-linear transport treatment.

For one illustrative lossless symmetric double barrier with single-barrier probabilities $T_b = 2/3$ and $R_b = 1/3$, summing the coherent multiple-reflection amplitudes gives $T(\phi) = T_b^2/|1 - R_b e^{i\phi}|^2 = 1/[1 + 3\sin^2(\phi/2)]$. The phase ϕ includes propagation and reflection contributions. This describes a model, not a measured device.

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Two phase points, opposite pressure responses

Assume, only for this example, that compression changes phase by $+\pi/6$ without changing the barriers. At $\phi = \pi/2$, T changes from $2/5$ to $4/13$; hence $\Delta G/G = -3/13 \approx -23.1\%$. At $\phi = 3\pi/2$, the same phase increment changes T from $2/5$ to $4/7$, giving $\Delta G/G = 3/7 \approx +42.9\%$.

The assumed phase shift is not a universal pressure law. Inelastic/dephasing processes, thermal averaging and several channels with different phases can weaken or alter the oscillation; other pressure-driven changes in barriers also violate the fixed-barrier model. An observed sign alone does not identify a unique mechanism.

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From observation to defensible claim

For every claim, attach (1) the measurement and sample preparation, (2) spatial and temporal resolution, (3) operating protocol and controls, and (4) competing interpretations.

Example: **Observed:** low-bias current map changes at a location after a pulse. **Supported:** a local electrical response changed under that protocol. **Not yet established:** that the spot is an isolated metal filament of the map's measured width. Composition and geometry require additional, matched evidence.

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A practical experiment checklist

Stage	Questions to record
Before	Device stack and state; tip condition; alignment; environment; baseline trace.
During	Bias, compliance, current, force, acquisition time, clock synchronization and dose.
After	Repeatability, control regions, device survival, registration, uncertainty and alternative explanations.

Use the checklist to decide which follow-up experiment would distinguish two surviving mechanisms, rather than treating a visually persuasive image as a complete proof.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Material cases: evidence and mechanisms

Four case families connect switching physics to measurements: SiO_2 conductive-bridge RAM, TaO_x RESET statistics, HfO_2/Ti aging, and two- and three-terminal atomic switches. Values below are either explicitly identified as historical observations or stated as hypothetical teaching models.

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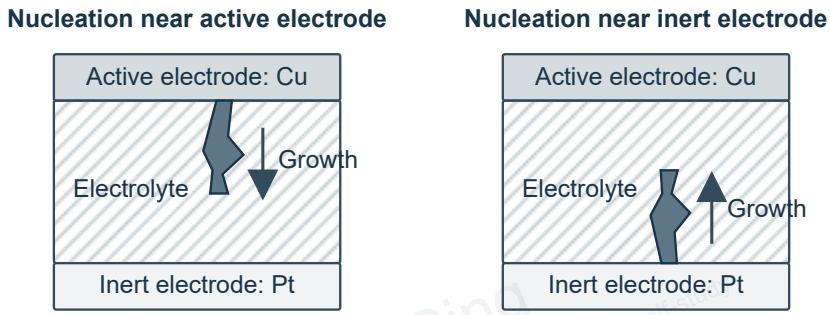
Compare mechanisms through their evidence

Case	Primary observation	A proposed explanation	What needs checking
SiO_2 CBRAM	switching and transport depend on bias and environment	metal ions participate in a conductive bridge	composition, polarity, transport and interfaces
TaO_x RESET	distributions differ with measurement and initial state	thermally assisted local rupture	current, temperature model and sample variability
HfO_2/Ti	the HRS/LRS window evolves during storage	oxygen redistribution and Ti oxidation	separate devices, aging protocol and depth-profile artifacts
Atomic switch	a conducting gap can be opened or closed	nucleation, growth and dissolution	field, pulse timing and alternative kinetics

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A conductive bridge is a process, not a single reaction

Conductive bridge: nucleation can differ
Schematic; reaction, supply and transport govern growth



Both scenarios are possible; polarity alone does not set direction.
Bridge formation also needs ion supply, reduction and connection.

For an illustrative Cu-active electrode, oxidation can supply ions: $\mathrm{Cu} \rightarrow \mathrm{Cu}^{+} + e^{-}$. A reduction step can return ions to metal: $\mathrm{Cu}^{+} + e^{-} \rightarrow \mathrm{Cu}$. Their locations and rates depend on electrode materials, interfacial kinetics, ion transport and electronic access.

Polarity alone does not determine morphology

Ion drift, diffusion, interfacial reduction and existing clusters can compete. Published material systems show growth toward either electrode or distributed nucleation; a voltage sign by itself cannot establish a universal growth direction. The return of HRS also need not remove every metallic atom.

To test a mechanism, align electrical switching with composition-sensitive imaging, repeat cycles, control humidity and temperature, and check whether the probe changes the state.

A diffusion scale is not a SET-time prediction

For ideal one-dimensional unbiased diffusion from a point, the mean-square displacement is $\langle x^2 \rangle = 2Dt$. For a deliberately hypothetical $D = 10^{-21}\,\mathrm{cm^2/s}$ and rms displacement $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$,

$$t = \frac{(2 \times 10^{-6})^2}{2 \times 10^{-21}} = 2 \times 10^{9}\,\mathrm{s} \approx 63\,\mathrm{yr}.$$

This extrapolation excludes electric-field drift, interfacial reactions and thin-film pathways. It is not a device switching time or evidence identifying one mechanism.

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Field and humidity change the boundary conditions

Under a field, ionic flux has drift and diffusion contributions; a high-field empirical fit applies only within its measured range. Boundary reaction rates and available ions also matter.

In a particular reported Cu/SiO_2/Pt experiment, changing water-vapor conditions affected the redox and ion-population interpretation; this does not establish that humidity always increases ionic mobility. Compare matched samples and histories before attributing a change to one parameter.

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Field drift: a limited numerical comparison

For a hypothetical dilute, isothermal, singly charged carrier in the linear-response regime, use the Einstein relation $\mu = D/(k_B T/e)$ and drift speed $v_d = \mu E$. Take $D = 10^{-21}\,\mathrm{cm^2/s}$, $k_B T/e = 0.02585\,\mathrm{V}$, and $E = 10^5\,\mathrm{V/cm}$:

$$\mu = 3.868 \times 10^{-20}\,\mathrm{cm^2/(V\,s)}, \quad v_d = 3.868 \times 10^{-15}\,\mathrm{cm/s}.$$

The drift distance divided by v_d for $L = 20\,\mathrm{nm}$ is $5.17 \times 10^8\,\mathrm{s}$. This is an illustrative drift scale, not a first-passage prediction or a SET time. Linear response and the Einstein relation are explicit assumptions; the field value by itself does not establish them. For a hopping picture, $qaE \ll k_B T$ is one condition to check. Constant field, dilute independent carriers and these relations may fail in a real switching gap.

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What changing humidity can and cannot identify

In a uniform drift-only illustrative model $J = qn\mu E$. The same change in J could result from a change in mobile-carrier concentration n, mobility μ, or field E; interface reactions and diffusive flux add further ambiguity. A current change alone cannot separately identify n and μ.

At fixed geometry and temperature, measure a concentration-sensitive signal and independently test transport across several fields and matched humidity histories; the result remains conditional on the chosen transport and contact model.

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A simple impedance model and its limits

Suppose a resistance R and capacitance C are in parallel. Their admittance is $Y = 1/R + j\omega C$; hence

$$Z = \frac{R}{1 + j\omega RC}, \qquad -\operatorname{Im} Z = \frac{\omega R^2 C}{1 + (\omega RC)^2}.$$

The imaginary component reaches its maximum at $\omega RC = 1$. Fitting this equivalent circuit describes a response under its assumptions; it does not uniquely reveal a filament's shape or composition.

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Work through the RC scale

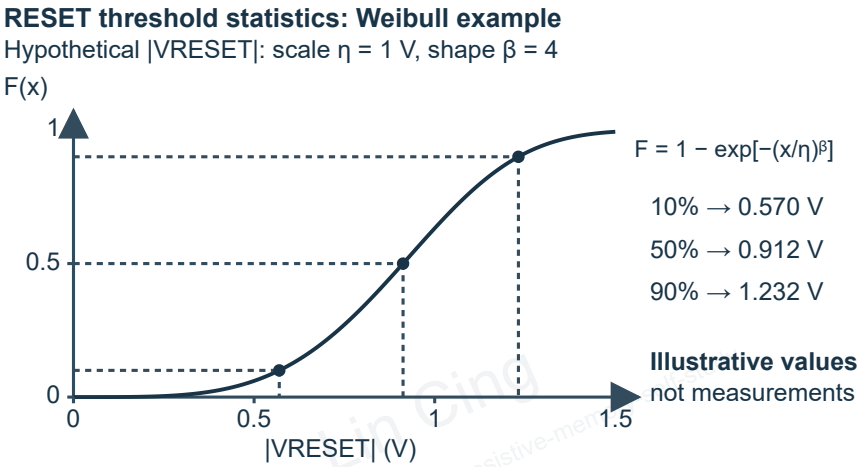
Let $R = 1\,\mathrm{M\Omega}$ and $C = 10\,\mathrm{pF}$ in the parallel model. Then $\tau = RC = 10\,\mu\mathrm{s}$ and

$$f_{\mathrm{peak}} = \frac{1}{2\pi RC} \approx 15.9\,\mathrm{kHz}.$$

Measure over a sufficient frequency range and check lead and electrode contributions. More than one physical arrangement may fit the same spectrum.

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RESET is a distribution, not one universal number



η: 63.2% quantile. Report binning, correlation and censoring.

One historical Ta/TaOx/Pt study used an approximately 18 nm oxide and repeated quasi-DC cycles on a device. SET compliance, RESET stop voltage, voltage polarity, read voltage and cycle history are part of that particular dataset, not universal settings.

For a negative-polarity RESET experiment, define the nonnegative variable $X = |V_{\mathrm{RESET}}|$ before fitting. A peak-current criterion or another chosen criterion must be stated explicitly.

One TaOx RESET study: protocol before inference

In the specific historical Ta/TaOx/Pt experiment, the oxide was about 18 nm; approximately 1,000 quasi-DC cycles were recorded on the same device. SET used positive bias with $4\,\mathrm{mA}$ compliance, RESET swept negative to a $-1.5\,\mathrm{V}$ stop, and R_{ON} was read at $0.1\,\mathrm{V}$; the voltage-step time was $1\,\mathrm{ms}$. RESET voltage was defined through the current peak.

These settings and the chosen event definition describe that dataset. Cycle history and resistance binning affect the observed distribution and are not independent population statistics.

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Six resistance bins under a simple assumption

Assume six hypothetical resistance bins and a common peak magnitude $|V_{\mathrm{RESET}}| = 0.9\,\mathrm{V}$. If the instantaneous peak is approximated by $I_{\mathrm{peak}} = |V|/R_{\mathrm{ON}}$:

R_{ON} (Ω)	100	150	200	250	300	400
I_{peak} (mA)	9	6	4.5	3.6	3	2.25

For example, $0.9/150 = 0.006\,\mathrm{A} = 6\,\mathrm{mA}$. These hypothetical fixed-voltage, Ohmic values are not a reconstruction of the published Weibull distribution. In this model the current is inversely proportional to resistance; evolving resistance or a nonconstant peak voltage invalidates the simple calculation.

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Weibull model: CDF and quantile

For $x \geq 0$, shape $\beta > 0$, and scale $\eta > 0$, a two-parameter Weibull model is

$$F(x) = 1 - e^{-(x/\eta)^{\beta}}, \qquad x_p = \eta[-\ln(1-p)]^{1/\beta}.$$

At $x = \eta$, $F(\eta) = 1 - e^{-1} \approx 0.632$. This is a model definition; fitting data does not establish that all devices obey it. Use magnitude consistently and treat censored observations explicitly.

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Weibull worked example

Assume $\eta = 1\,\mathrm{V}$, $\beta = 4$. At $0.8\,\mathrm{V}$, $(0.8)^4 = 0.4096$ and $F = 1 - e^{-0.4096} \approx 0.3361$.

The 10%, 50%, and 90% quantiles are approximately 0.5697, 0.9124, and $1.2318\,\mathrm{V}$. They describe the assumed distribution, not a measured performance guarantee.

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How to fit without manufacturing certainty

For $x > 0$ and $0 < F < 1$, $\ln[-\ln(1 - F)] = \beta \ln x - \beta \ln \eta$. A Weibull plot can reveal departures from a straight line; it cannot validate a mechanism on its own.

Report the sample count, device count, cycle dependence, selection or resistance binning, censoring and confidence intervals. Repeated cycles from one device are not automatically independent devices.

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A thermal RESET model is conditional

Assume steady local heating with constant R and thermal resistance R_{th} to a bath at T_0:

$$P = \frac{V^2}{R}, \qquad T = T_0 + R_{\mathrm{th}} P.$$

This lumped model excludes thermal transients, spatial temperature gradients, evolving resistance and nonthermal mechanisms. Agreement with it can support a hypothesis but cannot uniquely establish it.

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Thermal estimate and scaling

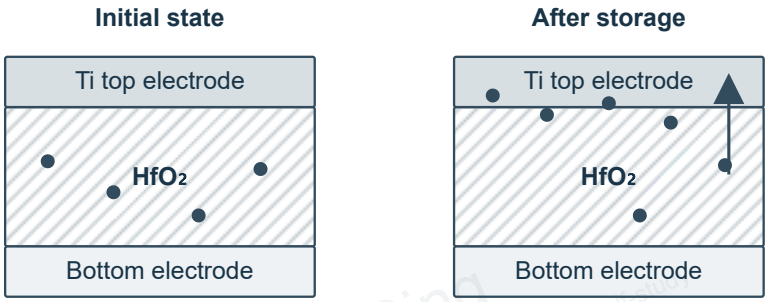
Take hypothetical $R = 150\,\Omega$, $V = 0.9\,\mathrm{V}$, $R_{\mathrm{th}} = 3 \times 10^4\,\mathrm{K/W}$. Then $P = 0.81/150 = 5.4\,\mathrm{mW}$ and $\Delta T = 162\,\mathrm{K}$.

If RESET requires the same critical temperature rise, $V_{\mathrm{RESET}}^2 = \Delta T_c R / R_{\mathrm{th}}$. A nearly constant RESET voltage across different R follows only if ΔT_c and R/R_{th} remain approximately fixed.

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Aging: a measurement protocol is part of the result

HfO_2/Ti ageing: material distribution and evidence
Conceptual illustration; not a measured concentration profile



- Oxygen-related species; arrow shows one possible change.

XPS is indirect; sputtering and sample variation affect interpretation.
Report storage time, structure and measurement conditions.

The cited HfO_2/Ti work studied Pt/HfO_2/Ti/Pt stacks and compared nominal Ti thicknesses, aging intervals and electrical states. The published curves concern specified batches and read conditions; they are not a lifetime guarantee for all HfO_2 cells.

Interpret a changing memory window

For a specified group, a reported HRS/LRS ratio fell from roughly 10^4 to 10^2 over about two weeks: the ratio is 100 times smaller. That does not determine the absolute HRS or LRS values individually.

Depth-profile XPS can support changes in elemental or chemical distributions, but sputtering and sample preparation may alter the profile. It is not a real-time movie of oxygen motion in the same operating cell.

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An aging comparison must preserve the sample identity

The reported stack was Pt/HfO_2 (about 10 nm)/Ti/Pt. Compare nominal Ti thicknesses 10, 30 and 100 nm; the approximately two-week fall from 10^4 to 10^2 concerns the 30 nm group under its particular protocol. The study first applied 15 cycles and then averaged five cycles per cell; storage comparisons used separate fresh devices from the same batch.

Step	What is compared	What the comparison does not establish
Prepare	Same-batch devices; specified Ti thickness and storage time	One device followed continuously unless explicitly measured
Condition and read	Forming and cycles under specified voltage/compliance and read convention	Equal initial states or identical HRS and LRS separately
Profile	Day 1 versus day 60 XPS depth profiles	Undisturbed, real-time motion during switching

At about 60 days, reported ratios for nominal 10/30/100 nm groups were approximately $5 \times 10^4/10^2/10$, respectively, with device and protocol dependence. Read conventions and some labels differ across adjacent source descriptions; do not merge ambiguous conditions without checking the original experiment.

Keep observations and explanations separate

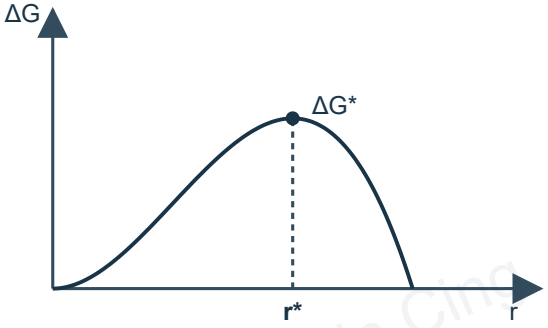
Compare matched fresh and aged devices, specify the Ti thickness, environment, preconditioning, read polarity and uncertainty. A TiO_2 insertion layer is a distinct device structure, not a continuation of a bare-Ti sample.

Reported figure labels and adjacent descriptions may differ in thickness or read conventions. Do not merge uncertain conditions: check the original experiment before reproducing or comparing its numeric values.

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A nucleation model for an atomic switch

Heterogeneous nucleation: surface versus volume energy
Model: fixed interfacial energy, driving force and cap shape



Small: surface cost
Large: volume gain

$r^* = 2\gamma / |\Delta g_v|$

$\Delta G^* = f(\theta)\, 16\pi\gamma^3 / [3|\Delta g_v|^2]$

Barrier ≠ rate: prefactors and transport also matter.
Nanoscale clusters or dynamic fields can invalidate the model.

In a classical spherical-cap approximation, let $\gamma > 0$ be surface energy, $|\Delta g_v| > 0$ the magnitude of bulk driving free energy per volume, and $0 < f(\theta) \leq 1$ a geometric factor. Then

$$\Delta G(r) = f(\theta)\left(4\pi r^2\gamma - \frac{4\pi}{3}r^3|\Delta g_v|\right).$$

Derive the critical radius and barrier

For fixed $f(\theta) > 0$, differentiate with respect to r: $d\Delta G/dr = f(8\pi r\gamma - 4\pi r^2|\Delta g_v|)$. The nonzero stationary radius is

$$r^* = \frac{2\gamma}{|\Delta g_v|}, \qquad \Delta G^* = f(\theta)\frac{16\pi\gamma^3}{3|\Delta g_v|^2}.$$

It is a maximum under this model. Classical continuous geometry and constant material properties may be poor approximations for a cluster only a few atoms wide.

Nucleation example: calculate before interpreting

Assume $\gamma = 0.03\,\mathrm{J/m^2}$, $|\Delta g_v| = 3 \times 10^7\,\mathrm{J/m^3}$ and $f = 0.1$. Then $r^* = 2 \times 10^{-9}\,\mathrm{m} = 2\,\mathrm{nm}$ and $\Delta G^* \approx 5.03 \times 10^{-20}\,\mathrm{J} \approx 0.314\,\mathrm{eV}$.

At 300 K, use $k_B T \approx 0.02585\,\mathrm{eV}$ and $e^{-\Delta G^*/k_B T} \approx 5.3 \times 10^{-6}$. This factor alone is not a nucleation rate: the attempt frequency, available sites and nonequilibrium transport are not given.

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Time-resolved switching needs a defined event

A voltage edge applied to a device and its parasitic capacitance can produce $I_C = C\,dV/dt$ before a conducting bridge forms. A later jump in conduction current is a different event.

Define SET time relative to the applied pulse and a stated onset criterion; bandwidth, trigger delay and capacitance affect the measurement. A published nanosecond value belongs to its particular structure and measurement chain.

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Separate a capacitive peak from a conducting event

Assume $C = 0.2\,\mathrm{pF}$ and $dV/dt = 1\,\mathrm{V/ns}$. Before conduction, the ideal capacitive current is $0.2\,\mathrm{mA}$; seeing this at $0.5\,\mathrm{ns}$ does not establish SET.

If the stated onset of conduction occurs at $1.2\,\mathrm{ns}$ after pulse start, the defined SET time is $1.2\,\mathrm{ns}$. Device energy requires $\int V(t) I_{\mathrm{device}}(t)\,dt$ with a defined interval and treatment of parasitic energy; a peak alone is insufficient.

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A slope that does not identify a mechanism

Consider two deliberately simplified models on the voltage window 1–$2\,\mathrm{V}$. A nucleation ansatz has $t = A\exp[\Delta G(V)/(k_B T)]$, with $\Delta G(V) = \Delta G_0 - 0.05\,\mathrm{eV}\,(V/1\,\mathrm{V})$, $k_B T = 0.025\,\mathrm{eV}$, and $\Delta G_0 > 0.10\,\mathrm{eV}$ so the barrier remains positive throughout this interval. Choose the combined prefactor so that $\ln(t/\mathrm{ns}) = 6 - 2(V/1\,\mathrm{V})$.

Alternatively, an assumed high-field migration law $v = v_0 \exp[eaV/(2k_B T L)]$, with $a = 0.3\,\mathrm{nm}$, $L = 3\,\mathrm{nm}$, and $L/v_0 = e^6\,\mathrm{ns}$, gives the same expression for $t = L/v$. Each law has its own assumptions; neither is established by the example.

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Evaluate competing timing models

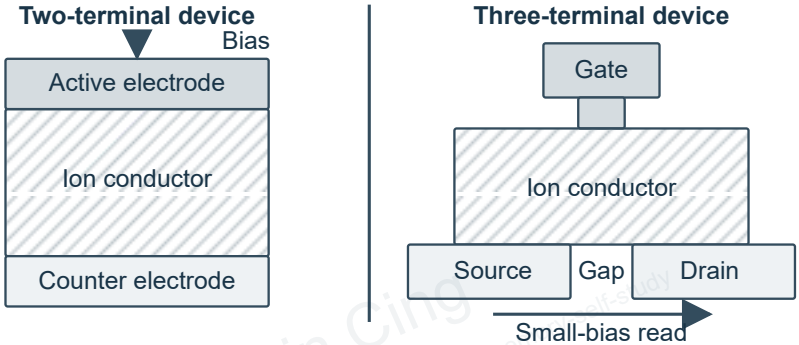
For either assumed law, $V = 1$, 1.5, and $2\,\mathrm{V}$ yield $t = e^4$, e^3, and $e^2\,\mathrm{ns}$: approximately 54.6, 20.1, and $7.39\,\mathrm{ns}$.

The identical fitted slope in this constructed example cannot distinguish mechanisms. Compare temperature, gap thickness, composition and time-resolved electrical/structural observations while considering model validity, parasitics and parameter uncertainty. Do not extrapolate a limited high-field fit to zero bias.

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Two terminals and three terminals answer different questions

Atomic switches: two-terminal and three-terminal
Schematics; materials and bias depend on the device



Connect source–drain while avoiding a gate short.
Ion motion and nucleation depend on material, bias and interfaces.

A two-terminal switch shares the applied voltage between the active electrode, electrolyte and counterelectrode. In a three-terminal example, a gate controls metal-ion supply while a smaller source–drain bias probes the gap. The conducting source–drain path must be distinguished from an unwanted short to gate.

Do not confuse a fit with a unique mechanism

Different kinetic models can fit a limited voltage dependence of switching time. Check their parameter values, geometry, temperature, pulse bandwidth and sensitivity to alternative assumptions. A fitted effective hopping distance larger than an atomic spacing may challenge one model, but a slope alone does not prove the remaining model.

As a final check, separate observed waveform, fitted model, inferred material change and untested predictions.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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